

Choose the correct relational algebra expressions to find all customers who bought either an umbrella or a raincoat.

- ☐ $\Pi_{\text{cust_id}}(\sigma_{\text{item}='umbrella'}(\text{Customer})) \cup \Pi_{\text{cust_id}}(\sigma_{\text{item}='raincoat'}(\text{Customer}))$
- ☐ $\Pi_{\text{cust_id}}(\sigma_{\text{item}='umbrella'}(\text{Customer})) \cap \Pi_{\text{cust_id}}(\sigma_{\text{item}='raincoat'}(\text{Customer}))$
- ☐ $\Pi_{\text{cust_id}}(\sigma_{\text{item}='umbrella' \vee \text{item}='raincoat'}(\text{Customer}))$
- ☐ $\Pi_{\text{cust_id}}(\sigma_{\text{item}='umbrella' \wedge \text{item}='raincoat'}(\text{Customer}))$

If E and F are two relational algebra expressions, then which among the following is NOT a valid relational algebra expression? **2 points**

- ☐ $E \cup F$
- ☐ $E \cap F$
- ☐ $E - F$
- ☐ $\sigma_p(E)$, where p is a predicate on attributes in E.
- ☐ $E + F$

Answer questions 3 to 6 based on relations R_1, R_2, R_3 given in Figure 1.

A	B
a ₁	b ₁
a ₁	b ₂
a ₂	b ₁
a ₃	b ₂
a ₁	b ₃
a ₂	b ₃

Relation R_1

B
b ₁
b ₂
b ₃

Relation R_2

A	B
a ₁	b ₂
a ₁	b ₁
a ₂	b ₁
a ₃	b ₂
a ₂	b ₂

Relation R_3

Figure 1: Relations R_1, R_2, R_3

ation to answer

Choose the relation that results from $R_1 \cap R_3$.

1 point

- ☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₂	b ₁
a ₃	b ₂

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₂	b ₂
a ₂	b ₃
a ₃	b ₂

☐

A	B
a ₂	b ₃

☐

A
a ₁

Choose the relation that results from $R_1 \cup R_3$.

1 point

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₃	b ₂

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₂	b ₂
a ₂	b ₃
a ₃	b ₂

☐

A	B
a ₂	b ₃

☐

A
a ₁

Choose the relation that results from $R_1 - R_3$.

1 point

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₃	b ₂

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₂	b ₂
a ₂	b ₃
a ₃	b ₂

☐

A	B
a ₁	b ₃
a ₂	b ₃

☐

A
a ₁

Choose the relation that results from $R_1 \div R_2$.

1 point

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₃	b ₂

☐

A	B
a ₁	b ₁
a ₁	b ₂
a ₁	b ₃
a ₂	b ₁
a ₂	b ₂
a ₂	b ₃
a ₃	b ₂

☐

A	B
a ₂	b ₃

☐

A
a ₁

Answer questions 7 to 10 based on relations **Students** and **Activity** given in Figure 2.

Name	Age	Marks	Subject
David	23	78	Maths
Matthew	29	54	English
Anand	22	89	JAVA
Mitchel	21	56	Maths
Shaun	26	92	Maths

Relation **Students**

Name	Sports	Awards	Points
David	Cricket	2	67
Matthew	Football	4	90
Anand	Cricket	5	80
Mitchel	Tennis	8	70
Shaun	Hockey	3	75

Relation **Activity**

Figure 2: Relations **Students** and **Activity**

Which of the following expressions will return the names of students whose age is greater than 25, or who are enrolled in Maths? **2 points**

- ☐ $\Pi_{\text{Name}}(\sigma_{\text{Age}<25 \vee \text{Subject}=\text{'Maths'}}(\text{Students}))$
- ☐ $\sigma_{\text{Name}}(\Pi_{\text{Age}>25 \vee \text{Subject}=\text{'Maths'}}(\text{Students}))$
- ☐ $\sigma_{\text{Name}}(\sigma_{\text{Age}>25 \wedge \text{Subject}=\text{'Maths'}}(\text{Students}))$
- ☐ $\Pi_{\text{Name}}(\sigma_{\text{Age}>25 \vee \text{Subject}=\text{'Maths'}}(\text{Students}))$

Consider the relational algebra expression shown below:

2 points

$\Pi_{\text{Name}, \text{Sports}}(\sigma_{\text{Age}<25 \wedge \text{Marks}>75}(\text{Students} \bowtie \text{Activity}))$

What will be the output of the given expression?

☐

Name	Sports
David	Cricket
Matthew	Football
Anand	Cricket
Mitchel	Tennis
Shaun	Hockey

☐

Name	Sports
David	Cricket
Anand	Cricket

☐

Age	Marks
23	78
22	89

☐ None of the above

Consider the relational algebra expression shown below:

2 points

$\Pi_{\text{Name, Sports}}(\sigma_{\text{age} < 25 \wedge \text{Awards} > 3}(\text{Students} \bowtie \text{Activity}))$

Which among the following statements can be inferred from the given expression?

- ☐ It will find the Cartesian product of Students and Activity tables.
- ☐ It will return the names of students and sports played by the students whose age is less than 25 and have won more than 3 awards.
- ☐ It will return only the names of students whose age is more than 25 and have won more than 3 awards in a particular sports.
- ☐ Project (Π) operator is used to project the name and sports from the table obtained by ($\text{Students} \bowtie \text{Activity}$) based on ($\sigma_{\text{age} < 25 \wedge \text{Awards} > 3}$).

Choose the correct expression, which will result in the table shown below:

2 points

Name	Age	Sports
David	23	Cricket
Matthew	29	Football
Anand	22	Cricket
Mitchel	21	Tennis
Shaun	26	Hockey

- ☐ $\sigma_{\text{Name, Age}}(\text{Students}) \cap \sigma_{\text{Sports}}(\text{Activity})$
- ☐ $\Pi_{\text{Name, Age}}(\text{Students}) \cup \Pi_{\text{Sports}}(\text{Activity})$
- ☐ $\sigma_{\text{Name, Age}}(\text{Students}) \cup \sigma_{\text{Sports}}(\text{Activity})$
- ☐ $\Pi_{\text{Name, Age, Sports}}(\text{Students} \bowtie \text{Activity})$